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11.3.2 Jacobi's method for computing the Spectral Decomposition

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it affects from the right like that. Okay? Now, if you look at these entries right here, how these entries are affected by the actual Jacobi rotation that's embedded here is exactly this way. So if you take these entries and you move them into this matrix then you immediately see -- ah, if I pick my gamma and my sigma in such a way that when I act on that 2x2 matrix, I get a diagonal matrix, I have a means by which I can replace

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Reading assignment

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Read Unit 11.3.2 of the notes. [\[LINK\]](#)

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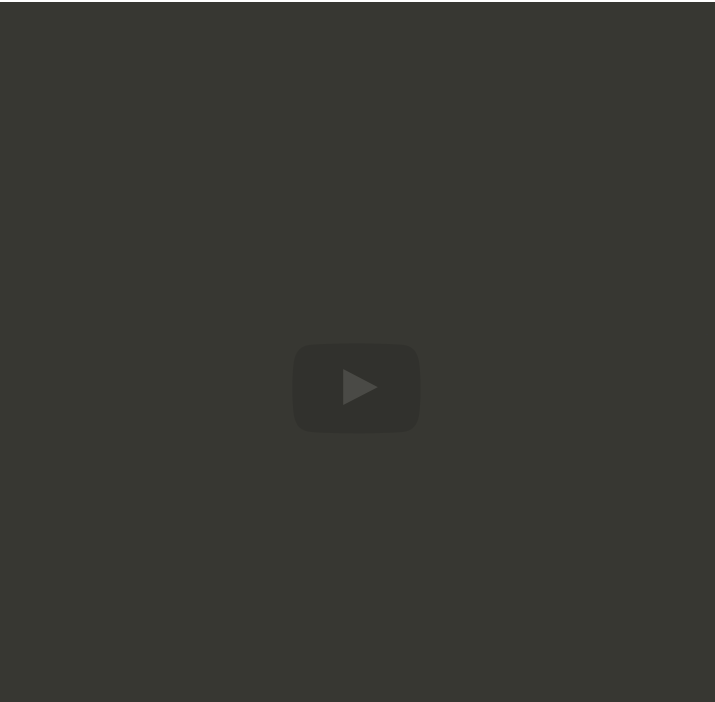
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performed by his student, Seidel. And you can imagine that Seidel would want to reduce the sum of the squares of the off-diagonal elements as much as possible for every time he does computation, because doing this by hand takes a lot of time. So what he -- and the other thing is they could only do this for very small matrices because, you know you're not going to do one hundred by hundred or a thousand by thousand

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Homework 11.3.2.1

1/1 point (graded)
Note: In the description of this homework, we index like Matlab does: starting at one. This is in contrast to how we usually index in these notes, starting at zero.

In `Assignments/Week11/matlab` , you will find the following files:

- `Jacobi_rotation.m` : A function that computes a Jacobi rotation from a 2×2 symmetric matrix.
- `Seidel.m` : A script that lets you apply Jacobi's method to a 5×5 matrix, much like Seidel did by hand. Fortunately, you only indicate the off-diagonal element to zero out. Matlab then does the rest.

Use this to gain insight into how Jacobi's method works. You will notice that finding the off-diagonal element that has largest magnitude is bothersome. You don't need to get it right every time.

Once you have found the diagonal matrix, restart the process. This time, zero out the off-diagonal elements in systematic "sweeps," zeroing the elements in the order.

```
( 2,1 ) - ( 3,1 ) - ( 4,1 ) - ( 5,1 ) -  
      ( 3,2 ) - ( 4,2 ) - ( 5,2 ) -  
            ( 4,3 ) - ( 5,3 ) -  
                  ( 5,4 )
```

and repeating this until convergence. A sweep zeroes every off-diagonal element exactly once (in symmetric pairs).

done



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Homework 11.3.2.2

1/1 point (graded)
Partition matrix A like

$$\begin{pmatrix} A_{00} & \textcolor{red}{a}_{10} & A_{20}^T & \textcolor{red}{a}_{30} & A_{40}^T \\ \textcolor{red}{a}_{10}^T & \textcolor{blue}{\alpha}_{11} & \textcolor{red}{a}_{21}^T & \textcolor{blue}{\alpha}_{31} & \textcolor{red}{a}_{41}^T \\ A_{20} & \textcolor{red}{a}_{21} & A_{22} & \textcolor{red}{a}_{32} & A_{42}^T \\ \textcolor{red}{a}_{30}^T & \textcolor{blue}{\alpha}_{31} & \textcolor{red}{a}_{32}^T & \textcolor{blue}{\alpha}_{33} & \textcolor{red}{a}_{43}^T \\ A_{40} & \textcolor{red}{a}_{41} & A_{42} & \textcolor{red}{a}_{43} & A_{44} \end{pmatrix}$$

and let J equal the Jacobi rotation that zeroes the element denoted with α_{31} :

$$\begin{aligned} J^T A J &= \begin{pmatrix} I & 0 & 0 & 0 & 0 \\ 0 & \textcolor{blue}{\gamma}_{11} & 0 & \textcolor{blue}{\sigma}_{31} & 0 \\ 0 & 0 & I & 0 & 0 \\ 0 & \textcolor{blue}{-\sigma}_{31} & 0 & \textcolor{blue}{\gamma}_{33} & 0 \\ 0 & 0 & 0 & 0 & I \end{pmatrix} \begin{pmatrix} A_{00} & \textcolor{red}{a}_{10} & A_{20}^T & \textcolor{red}{a}_{30} & A_{40}^T \\ \textcolor{red}{a}_{10}^T & \textcolor{blue}{\alpha}_{11} & \textcolor{red}{a}_{21}^T & \textcolor{blue}{\alpha}_{31} & \textcolor{red}{a}_{41}^T \\ A_{20} & \textcolor{red}{a}_{21} & A_{22} & \textcolor{red}{a}_{32} & A_{42}^T \\ \textcolor{red}{a}_{30}^T & \textcolor{blue}{\alpha}_{31} & \textcolor{red}{a}_{32}^T & \textcolor{blue}{\alpha}_{33} & \textcolor{red}{a}_{43}^T \\ A_{40} & \textcolor{red}{a}_{41} & A_{42} & \textcolor{red}{a}_{43} & A_{44} \end{pmatrix} \\ &= \begin{pmatrix} A_{00} & \textcolor{red}{\hat{a}}_{10} & A_{20}^T & \textcolor{red}{\hat{a}}_{30} & A_{40}^T \\ \textcolor{red}{\hat{a}}_{10}^T & \textcolor{blue}{\hat{\alpha}}_{11} & \textcolor{red}{\hat{a}}_{21}^T & 0 & \textcolor{red}{\hat{a}}_{41}^T \\ A_{20} & \textcolor{red}{\hat{a}}_{21} & A_{22} & \textcolor{red}{\hat{a}}_{32} & A_{42}^T \\ \textcolor{red}{\hat{a}}_{30}^T & 0 & \textcolor{red}{\hat{a}}_{32}^T & \textcolor{blue}{\hat{\alpha}}_{33} & \textcolor{red}{\hat{a}}_{43}^T \\ A_{40} & \textcolor{red}{\hat{a}}_{41} & A_{42} & \textcolor{red}{\hat{a}}_{43} & A_{44} \end{pmatrix} \end{aligned}$$

Show that $\|\text{off}(\hat{A})\|_F^2 = \|\text{off}(A)\|_F^2 - 2\alpha_{31}^2$.

done



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